F70LA Life Insurance Mathematics A (2024/25)
Tutorial 9 Solutions

1. (a) Let the random variable $T_{[60]}$ denote the future lifetime of ([60]). Let $h(T_{[60]})$ be the present value of the office's profit for the contract. Then at $i = 0.04$,

$$h(T_{[60]}) = 47\,\bar{a}_{\overline{T_{[60]}|}} - 1\,000\,v^{T_{[60]}} = 47\frac{1 - v^{T_{[60]}}}{\ln(1.04)} - 1\,000\,v^{T_{[60]}} = 1\,198.35 - 2\,198.35\,v^{T_{[60]}}.$$

(i) From the Tables, $A_{[60]} = 0.45510$ and using the claims acceleration approximation, $\bar{A}_{[60]} \approx (1.04)^{0.5} A_{[60]} = 0.46411$. Hence, as $\mathbb{E}\left(v^{T_{[60]}}\right) = \bar{A}_{[60]}$,

$$\mathbb{E}\left(h(T_{[60]})\right) = 1\,198.35 - 2\,198.35\,\bar{A}_{[60]} \approx 1\,198.35 - 2\,198.35\,(0.46411) = 178.07.$$

Hence the expected value of the present value of the office's profit on the contract is £178.07.

(ii) Calculating the variance of $h(T_{[60]})$,

$$\text{var}\left(h(T_{[60]})\right) = (2\,198.35)^2 \left(\bar{A}^{8.16\%}_{[60]} - \left(\bar{A}^{4\%}_{[60]}\right)^2\right).$$

From the Tables, $A^{8.16\%}_{[60]} = 0.23547$. The claims acceleration approximation then gives that

$$\bar{A}^{8.16\%}_{[60]} \approx (1.0816)^{0.5} A^{8.16\%}_{[60]} = (1.04)(0.23547) = 0.24489.$$

The variance of $h(T_{[60]})$ is

$$\text{var}\left(h(T_{[60]})\right) \approx (2\,198.35)^2 \left(0.24489 - (0.46411)^2\right) = (2\,198.35)^2 (0.17173)^2.$$

and hence the standard deviation of the present value of the office's profit on the contract is $(2\,198.35)(0.17173) = £377.52$.

(b) For the block of business,

$$\mathbb{E}\,(\text{Present value of total profit}) = 500(178.07) = £89\,035.00.$$

and

$$\text{sd}\,(\text{Present value of total profit}) = \sqrt{500}(377.52) = £8\,441.60.$$

2. (a) Using claims acceleration,

$$\bar{A}_{40:\overline{20|}} = \bar{A}^{1}_{40:\overline{20|}} + v^{20}\,{}_{20}p_{40} \approx (1+i)^{0.5} A^{1}_{40:\overline{20|}} + v^{20}\,{}_{20}p_{40} = (1+i)^{0.5}(A_{40:\overline{20|}} - v^{20}\,{}_{20}p_{40}) + v^{20}\,{}_{20}p_{40}.$$

At an effective interest rate of 4% per year and looking up the AM92 ultimate table, ${}_{20}p_{40} = \frac{9\,287.2164}{9\,856.2863} = 0.942263$. Also from the Tables, $A_{40:\overline{20|}} = 0.46423$. Hence

$$\bar{A}_{40:\overline{20|}} \approx (1.04)^{0.5}(0.46423 - (1.04)^{-20}\,(0.942263)) + (1.04)^{-20}(0.942263) = 0.46491.$$

(b)

$${}_{10|}\ddot{a}_{40:\overline{20|}} = {}_{10}p_{40}\, v^{10}\, \ddot{a}_{50:\overline{20|}} = {}_{10}p_{40}\, v^{10}\, (\ddot{a}_{50} - v^{20}\,{}_{20}p_{50}\, \ddot{a}_{70}).$$

From the Tables,

$$\ddot{a}_{50} = 17.444, \quad \ddot{a}_{70} = 10.375, \quad {}_{20}p_{50} = \frac{8\,054.0544}{9\,712.0728} = 0.829283, {}_{10}p_{40} = \frac{9\,712.0728}{9\,856.2863} = 0.985368,$$

and so

$${}_{10|}\ddot{a}_{40:\overline{20|}} = (0.985368)\,(1.04)^{-10}\,(17.444 - (1.04)^{-20}\,(0.829283)\,(10.375)) = 8.99821.$$

(c) The payment is 1 unit if death occurs in the first year, 2 units in the second, until the 20th year when the last payment is 20 units. Value the 20-year term insurance part of this increasing endowment insurance as the difference between two increasing whole life insurances. However, from $(IA)_{[40]}$, the payment is 21 units in the 21st year when the life is aged 60. To get 21 units from $(IA)_{60}$, which pays only 1 unit when the life is aged 60, we need to add on additional 20 units via $20A_{60}$. The same correction follows for when the life is aged 61, 62 and so on. Hence

$$(IA)_{[40]:\overline{20|}} = (IA)_{[40]} - v^{20}\,{}_{20}p_{[40]}\ ((IA)_{60} + 20A_{60}) + 20v^{20}\,{}_{20}p_{[40]}.$$

From the Tables,

$$(IA)_{[40]} = 7.95835, \quad (IA)_{60} = 8.36234, \quad A_{60} = 0.45640, \quad {}_{20}p_{[40]} = \frac{9\,287.2164}{9\,854.3036} = 0.942453$$

and so

$$\begin{aligned}(IA)_{[40]:\overline{20|}} =& 7.95835 - (1.04)^{-20}\,(0.942453)\ (8.36234 + 20(0.45640)) + 20(1.04)^{-20}\,(0.942453)\\ =& 0.435348 + 8.602465\\ =& 9.03788.\end{aligned}$$

3. Let P be the annual gross premium and let S be the sum insured. Then P is the solution to

$$0.98P\ddot{a}_{[45]:\overline{20|}} - 0.08P - 50 - 4\,\ddot{a}_{[45]:\overline{20|}} + 4 = S \cdot A_{[45]:\overline{20|}}.$$

Rearranging and using that, from the Tables, $\ddot{a}_{[45]:\overline{20|}} = 13.785$ and $A_{[45]:\overline{20|}} = 0.46982$,

$$P = \frac{46 + 4\,\ddot{a}_{[45]:\overline{20|}} + S \cdot A_{[45]:\overline{20|}}}{0.98\ddot{a}_{[45]:\overline{20|}} - 0.08} = \frac{46 + 4(13.785) + (0.46982)S}{0.98(13.785) - 0.08} = \frac{101.14 + (0.46982)S}{13.4293}.$$

If $S = £5\,000$ then $P = £182.45$. If $S = £10\,000$ then $P = £357.38$.

4. (a) Let the single premium be $P(\lambda)$. Then $P(\lambda) = \lambda \cdot P(\lambda) \cdot A^1_{30:\overline{35}|} + 1000\,{}_{35|}\ddot{a}_{30}$. Hence

$$P(\lambda) = \frac{1000 v^{35}\,{}_{35}p_{30}\,\ddot{a}_{65}}{1 - \lambda\,(A_{30:\overline{35}|} - v^{35}\,{}_{35}p_{30})}.$$

From the Tables, using AM92 ultimate and 4% per annum effective interest rate,

$${}_{35}p_{30} = \frac{8\,821.2612}{9\,925.2094} = 0.888773, \quad \ddot{a}_{65} = 12.276, \quad A_{30:\overline{35}|} = 0.26647.$$

Hence $P(\lambda) = \frac{2\,764.9091}{1-0.041241\lambda}$. Calculating $P(\lambda)$ gives the following values as λ varies.

λ	0	0.5	1
$P(\lambda)$	2 764.91	2 823.12	2 883.84

(b) The cost of providing a 100% return of premium if the would-be annuitant does not survive to get the annuity is relatively low, being about 4.3% of the cost of the deferred annuity without any return of premium benefit. This would be cheap benefit to offer to policyholders and give them a feeling of better value-for-money if they were to die before age 65.

(c) The value of the annuity benefit at age 65 is

$$\ddot{a}_{\overline{5}|} + {}_{5|}\ddot{a}_{65} = \ddot{a}_{\overline{5}|} + v^5\,{}_5p_{65}\,\ddot{a}_{70} = \frac{1-(1.04)^{-5}}{1-(1.04)^{-1}} + (1.04)^{-5}\,\frac{8\,054.0544}{8\,821.2612}(10.375) = 12.41573,$$

where the Tables were looked up for ℓ_{65}, ℓ_{70} and $\ddot{a}_{70}$.
Let the single premium for the new policy be $P'(\lambda)$. Then

$$P'(\lambda) = \frac{1000 v^{35}\,{}_{35}p_{30}\,\left(\ddot{a}_{\overline{5}|} + {}_{5|}\ddot{a}_{65}\right)}{1 - \lambda\,(A_{30:\overline{35}|} - v^{35}\,{}_{35}p_{30})} = \frac{2\,796.3803}{1 - 0.041241\lambda}.$$

Evaluating $P'(\lambda)$ gives the following values.

λ	0	0.5	1
$P'(\lambda)$	2 796.38	2 855.26	2 916.67

5. (a) $P = 20\,000\,a_{[60]}$. From the Tables, $\ddot{a}_{[60]} = 11.919$ at $i = 6\%$ per annum effective. Hence

$$P = 20\,000\,(\ddot{a}_{[60]} - 1) = 20\,000\,(10.919) = £218\,380.00$$

(b) The PV of the profit to the office is

$$\text{PV profit} = P - 20\,000\,a_{\overline{K_{[60]}|}} = 20\,000\left(10.919 - a_{\overline{K_{[60]}|}}\right) > 0 \quad \Leftrightarrow \quad a_{\overline{K_{[60]}|}} < 10.919.$$

Solving $a_{\overline{K_{[60]}|}} < 10.919$,

$$\frac{1 - v^{K_{[60]}}}{i} < 10.919 \quad \Rightarrow \quad K_{[60]} < -\frac{\ln(1 - 10.919(0.06))}{\ln(1.06)} = 18.3 \quad \Rightarrow \quad K_{[60]} \leq 18,$$

the last inequality following as $K_{[60]}$ takes only integer values. The probability that $K_{[60]} \leq 18$ is the probability that the life ([60]) dies before reaching age 79, i.e.

$$ {}_{19}q_{[60]} = 1 - \frac{\ell_{79}}{\ell_{[60]}} = 1 - \frac{5\,619.7577}{9\,263.1422} = 0.39332. $$

(c) As

$$ \text{PV profit} = 218\,380 - 20\,000 a_{\overline{K_{[60]}|}} = 218\,380 - 20\,000(\ddot{a}_{\overline{K_{[60]}+1|}} - 1), $$

we find

$$ \begin{aligned} \text{sd(PV profit)} = 20\,000 \cdot \text{sd}\left(\ddot{a}_{\overline{K_{[60]}+1|}}\right) =& \frac{20\,000}{d}\sqrt{({}^2A_{[60]} - (A_{[60]})^2)} \\ =& \frac{20\,000\sqrt{0.13907 - (0.32533)^2}}{1 - (1.06)^{-1}} \\ =& 64\,409.86. \end{aligned} $$

(d) Let a random variable Y be distributed as $Y \sim \mathcal{N}(\mu, \sigma^2)$ if Y is normally distributed with mean μ and variance σ.

Let X_n be a random variable representing the present value of the profit to the office from the nth policy, for $n = 1, 2, \ldots, 1\,000$. Let $K^n_{[60]}$ be a random variable representing the future curtate lifetime of the life on which the nth policy is written. Then $X_n = P - 20\,000\, a_{\overline{K^n_{[60]}|}}$. Then

$$ \mathbb{E}(X_n) = P - 20\,000\, a_{[60]} = P - 218\,380.00 $$

and

$$ \text{var}(X_n) = (20\,000)^2 \text{var}\left(\ddot{a}_{\overline{K^n_{[60]}+1|}}\right) = (64\,409.86)^2. $$

The assumption is that $K^1_{[60]}, K^2_{[60]}, \ldots$ are independent random variables and hence $X_1, X_2, \ldots$ are too, since they inherit the independence.

Let X be a random variable representing the present value of the profit to the office from 1 000 policies, i.e.

$$ X = \sum_{n=1}^{1\,000} X_n. $$

Since X is the sum of 1 000 independent random variables, we can apply the Central Limit Theorem to assert that X is normally distributed, with known mean and variance, i.e.

$$ X \sim \mathcal{N}\left(1000(P - 218\,380),\, 1000(64\,410)^2\right). $$

We want to calculate P satisfying $\alpha = \mathbb{P}[X > 0]$, for $\alpha \in \{0.95, 0.99\}$, i.e.

$$ \alpha = \mathbb{P}\left[\frac{X - 1000(P - 218\,380)}{64\,410\sqrt{1000}} > -\frac{1000(P - 218\,380)}{64\,410\sqrt{1000}}\right] = \mathbb{P}\left[Z > -\frac{1000(P - 218\,380)}{64\,410\sqrt{1000}}\right], $$

where $Z \sim \mathcal{N}(0,1)$. From the tables of the normal distribution,

$$\mathbb{P}[Z > -1.6449] = 0.95 \qquad \text{and} \qquad \mathbb{P}[Z > -2.3263] = 0.99.$$

Hence for $\alpha = 0.95$, we require

$$-\frac{1000(P - 218\,380)}{64\,410\sqrt{1000}} = -1.6449 \quad \Rightarrow \quad P = 218\,380 + (1.6449)\frac{64\,410}{\sqrt{1000}} = £221\,730.37$$

and for $\alpha = 0.99$, we require

$$-\frac{1000(P - 218\,380)}{64\,410\sqrt{1000}} = -2.3263 \quad \Rightarrow \quad P = 218\,380 + (2.3263)\frac{64\,410}{\sqrt{1000}} = £223\,118.26.$$

(e) Observe that the value of P increases with α, as we can see in part (d). The more certainty that is required about having a positive profit, the higher the premium charged.

Other comments:

i. For a fixed α, and defining z_α as the solution to $\mathbb{P}[Z > -z_\alpha] = \alpha$, observe that$P$ is a decreasing function of n, since

$$P = 218\,380 + z_\alpha \frac{64\,410}{\sqrt{n}} \to 218\,380 \quad \text{as } n \to \infty.$$

In other words, the more policies which are sold, the lower the variance of the average loss per policy. Individual policies will make profits or losses, but when we average these profits and losses across all of the sold policies, the average profit/loss declines to zero as the number of policies sold increases. The premium converges to the equivalence premium principle value, £218 380, as $n \to \infty$.

ii. We are assuming that the only source of uncertainty is the time of death of the lives (i.e. mortality). In particular, we are assuming that the interest rate is known and that the survival model, AM92 Ultimate, is known. The results above hold only if the interest rate and distribution of the times of death, given by AM92 Ultimate, are unchanged.